



Host a static website using AWS services

ZIUS KUMAR - 12310426 | Winter-pep | 31-3-26

Introduction

- This project demonstrates how to deploy and host a static website using Amazon Web Services (AWS). The workflow includes uploading files to Amazon S3, transferring them to an EC2 instance, and hosting the content over HTTP using a web server.
- The objective is to understand cloud storage, virtual servers, and deployment processes in AWS. This project provides hands-on experience with services like Amazon S3 and Amazon EC2.

OBJECTIVE

The objective of this project is to deploy and host a static website using AWS services by uploading files to Amazon S3, transferring them to an EC2 instance, and serving them over HTTP.

3. TOOLS & TECHNOLOGIES

-  AWS S3 (Storage)
-  AWS EC2 (Virtual Server)
-  Linux (Amazon Linux / Ubuntu)
-  Apache Web Server
-  AWS CLI

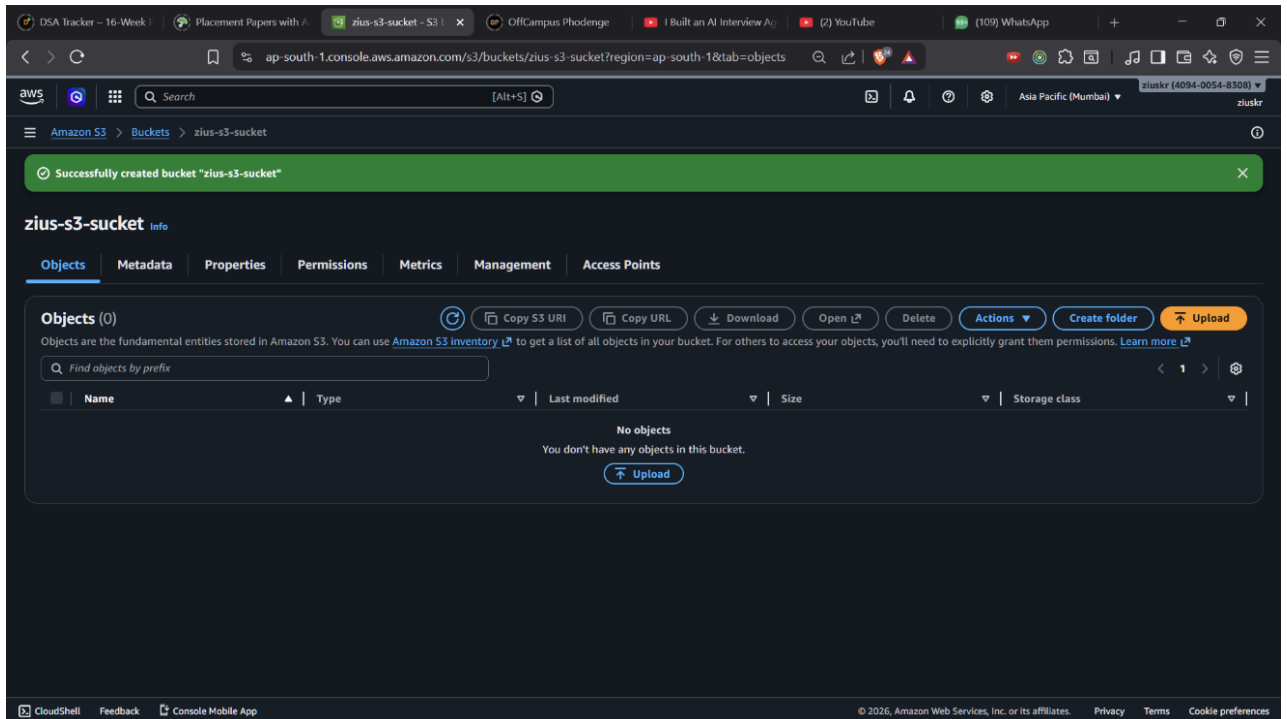
4. ARCHITECTURE

Local System → S3 Bucket → EC2 Instance → HTTP Hosting
(Browser)

5. STEP 1: CREATED S3 BUCKET

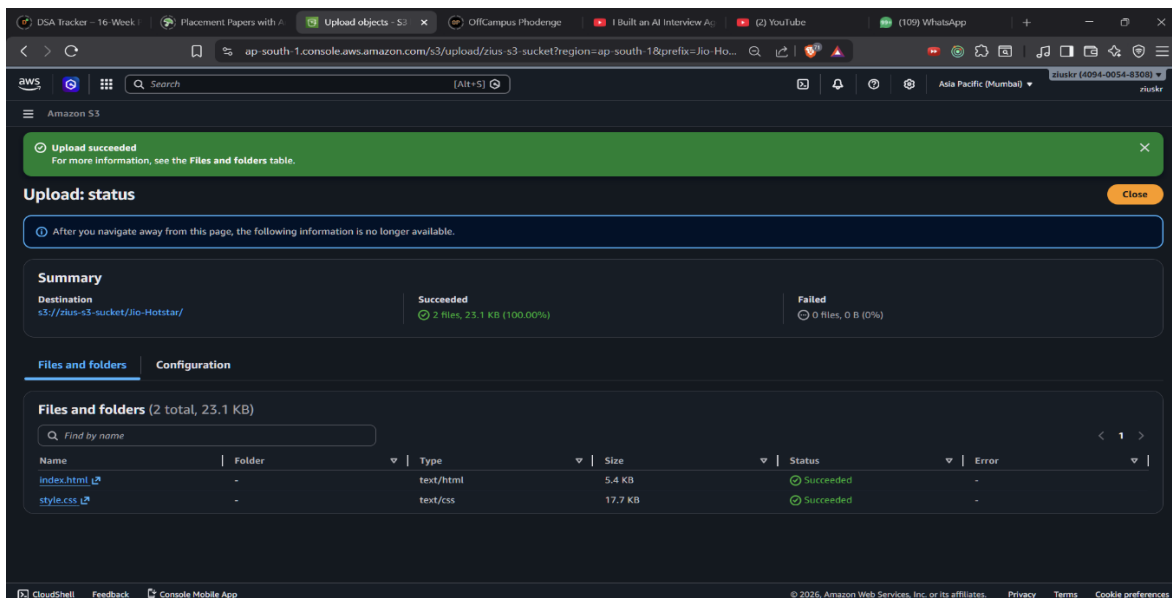
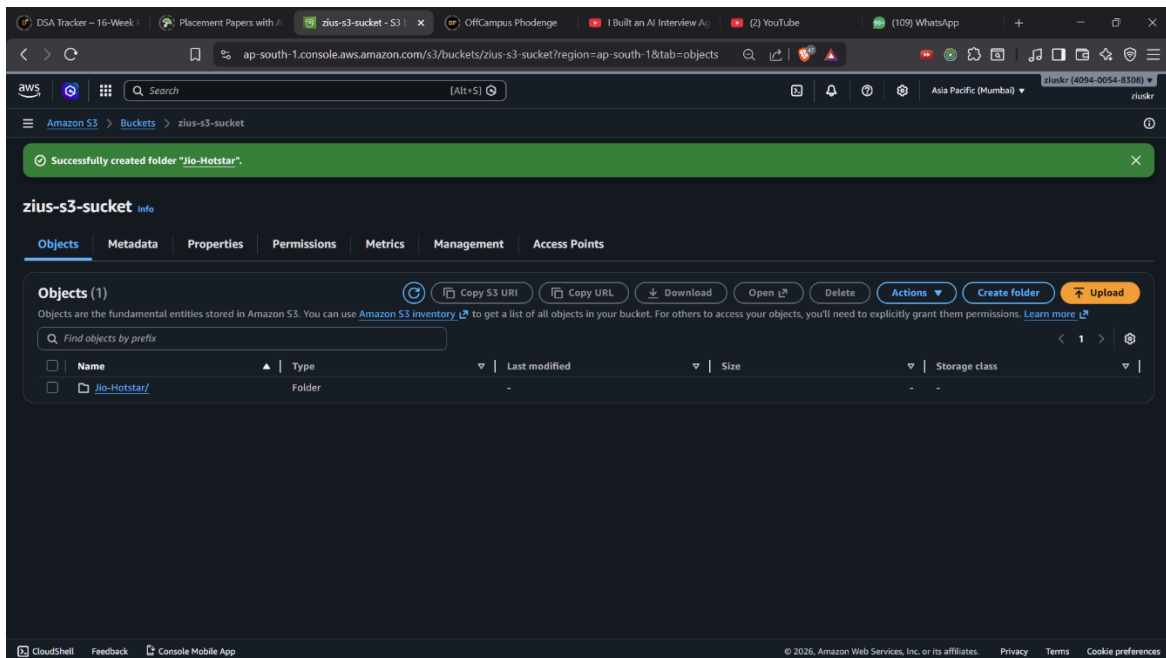
Local System → S3 Bucket → EC2 Instance → HTTP Hosting
(Browser)

- Created bucket in **ap-south-1 (Mumbai)**
- Named: **zius-s3-sucket**



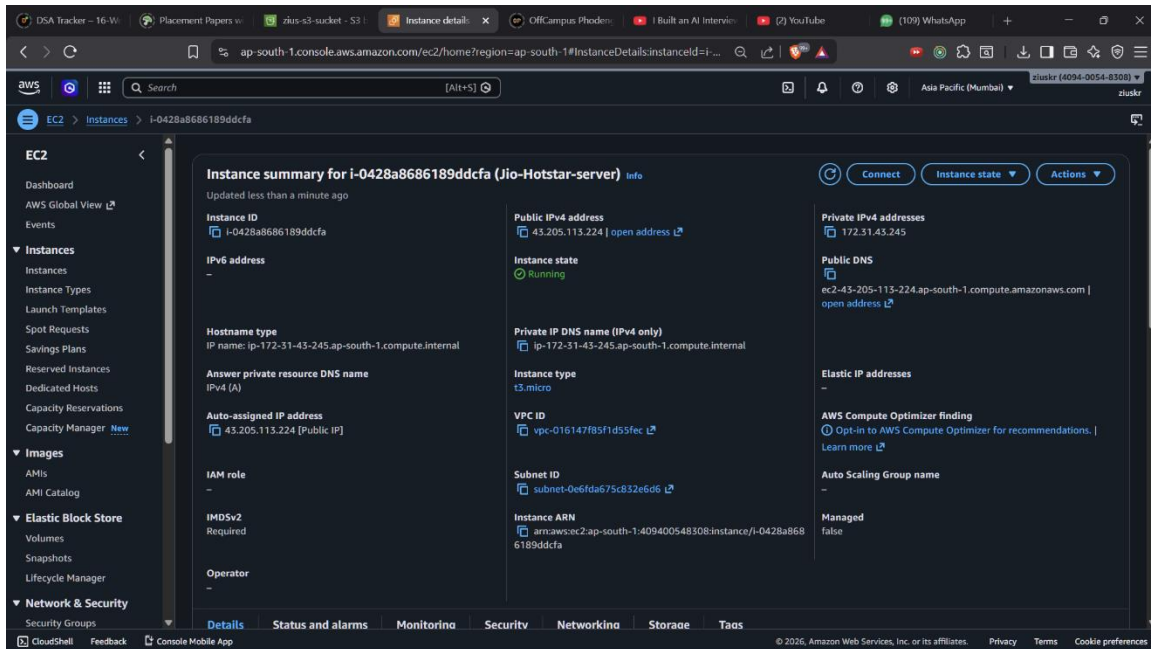
5. STEP 2: CREATED FOLDER & UPLOADED FILES

- Created folder: Jio-Hotstar
- Uploaded:
 - index.html
 - style.css
 - script.js



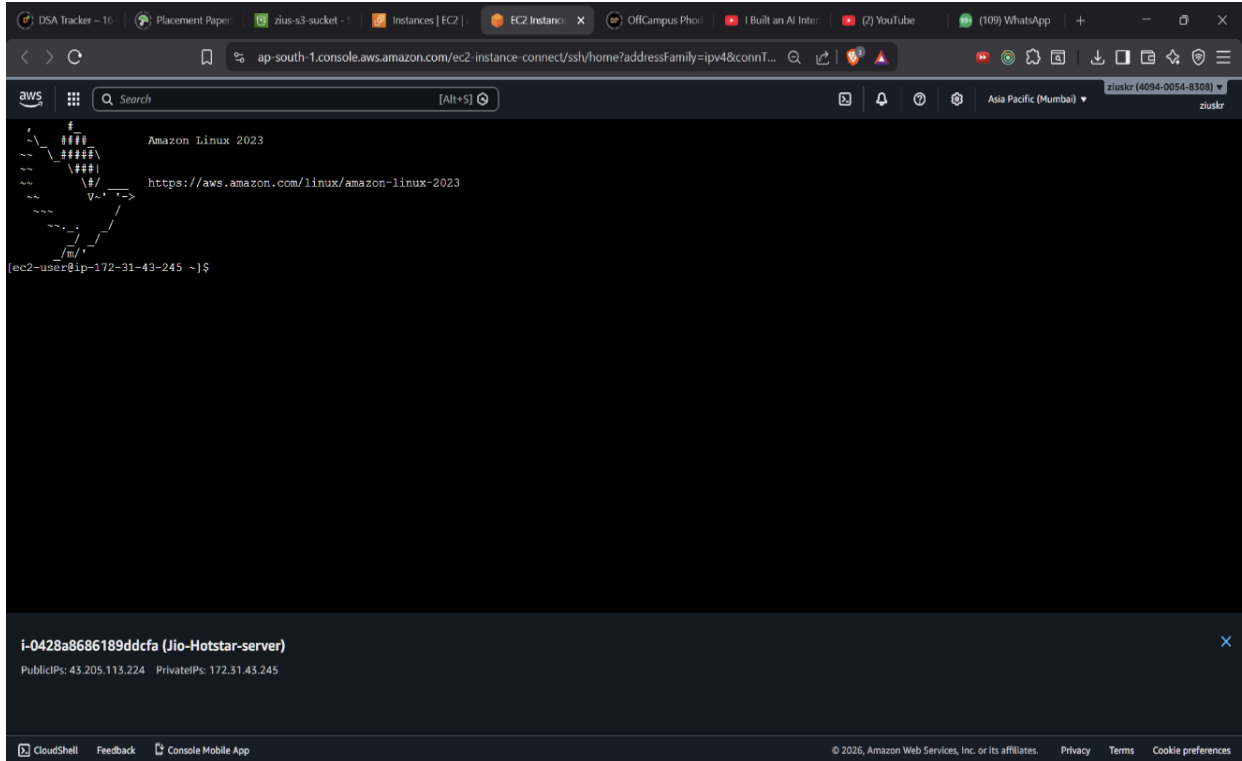
6. STEP 3: LAUNCHED EC2 INSTANCE

- OS: Amazon Linux
- Instance: t2.micro
- Enabled:
 - SSH (22)
 - HTTP (80)



7. STEP 4: CONNECTED TO EC₂

- Used SSH / EC₂ Instance Connect
- Opened termina



8. STEP 5: INSTALLED APACHE SERVER

Commands used:

```
sudo yum update -y
```

```
sudo yum install httpd -y
```

```
sudo systemctl start httpd
```

```
sudo systemctl enable httpd
```

Installed Apache

Started web server

Enabled auto-start

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-43-245 ~]$ sudo yum update -y
-bash: sudo: command not found
[ec2-user@ip-172-31-43-245 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-43-245 ~]$ sudo yum install httpd -y
Last metadata expiration check: 0:00:19 ago on Tue Mar 31 10:55:58 2026.
Dependencies resolved.

Package               Architecture      Version           Repository        Size
Installing:
httpd                  x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      47 k
Installing dependencies:
apr                    x86_64            1.7.5-1.amzn2023.0.4  amazonlinux      129 k
apr-util               x86_64            1.6.3-1.amzn2023.0.2  amazonlinux      97 k
apr-util-lmdb          x86_64            1.6.3-1.amzn2023.0.2  amazonlinux      13 k
generic-logos-httpd    x86_64            18.0.0-12.amzn2023.0.3 noarch           19 k
httpd-core             x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      1.4 k
httpd-filesystem       x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      13 k
httpd-tools            x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      81 k
libbrotli              x86_64            1.0.9-4.amzn2023.0.2  amazonlinux      315 k

i-0428a8686189ddcfa (Jio-Hotstar-server)
PublicIPs: 43.205.113.224 PrivateIPs: 172.31.43.245

Installing : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 3/13
Installing : apr-util-1.6.3-1.amzn2023.0.2.x86_64 4/13
Installing : mailcap-2.1.49-3.amzn2023.0.3.noarch 5/13
Installing : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 6/13
Installing : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 7/13
Running scriptlet: httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Installing : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Installing : httpd-core-2.4.66-1.amzn2023.0.1.x86_64 9/13
Installing : mod_httpd-2.0.27-1.amzn2023.0.3.x86_64 10/13
Installing : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 11/13
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 12/13
Installing : httpd-2.4.66-1.amzn2023.0.1.x86_64 13/13
Running scriptlet: httpd-2.4.66-1.amzn2023.0.1.x86_64 13/13
Verifying : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
Verifying : apr-util-1.6.3-1.amzn2023.0.2.x86_64 2/13
Verifying : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64 3/13
Verifying : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 4/13
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 5/13
Verifying : httpd-2.4.66-1.amzn2023.0.1.x86_64 6/13
Verifying : httpd-core-2.4.66-1.amzn2023.0.1.x86_64 7/13
Verifying : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Verifying : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 9/13
Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 10/13
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13
Verifying : mod_httpd-2.0.27-1.amzn2023.0.3.x86_64 12/13
Verifying : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 13/13

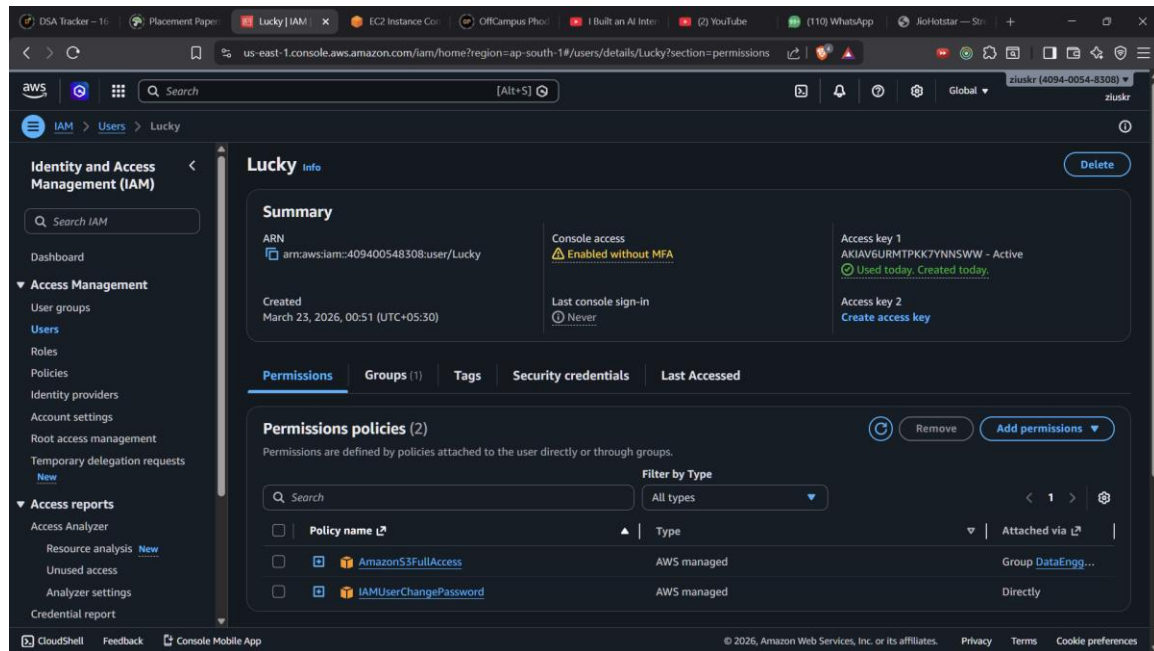
Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64 apr-util-1.6.3-1.amzn2023.0.2.x86_64 apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64 apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch httpd-2.4.66-1.amzn2023.0.1.x86_64 httpd-core-2.4.66-1.amzn2023.0.1.x86_64 httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 libbrotli-1.0.9-4.amzn2023.0.2.x86_64 mailcap-2.1.49-3.amzn2023.0.3.noarch mod_httpd-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-43-245 ~]$
```

9. STEP 6: CREATED IAM USER & ACCESS KEYS

- Created IAM user
- Generated:
 - Access Key ID
 - Secret Access Key

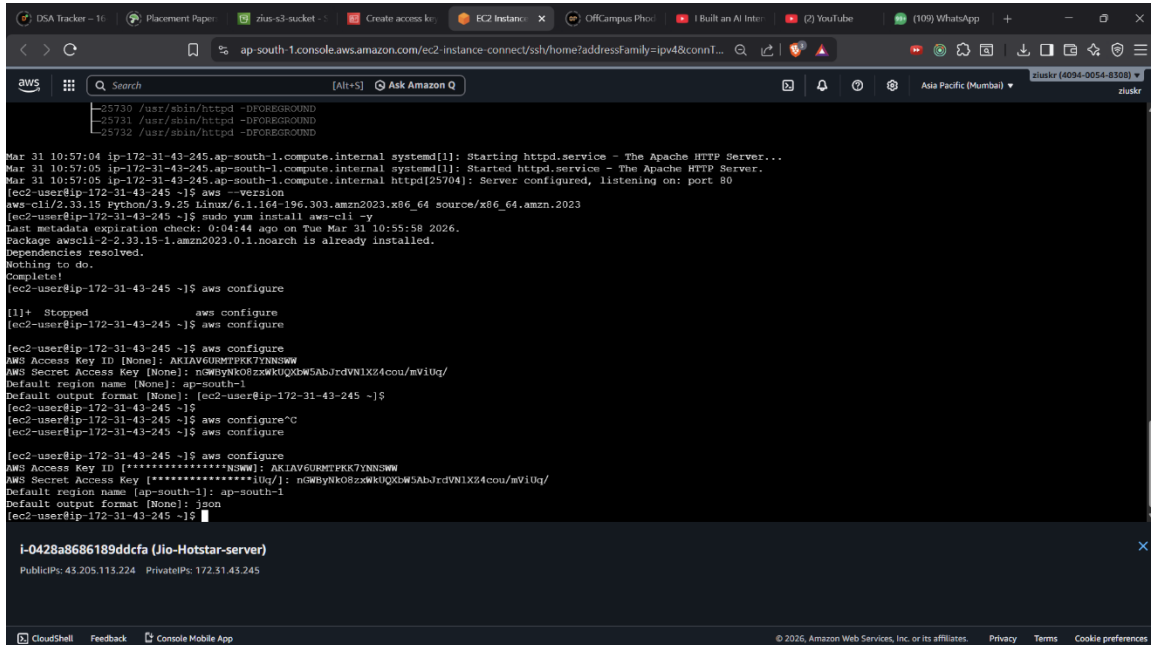
Used for AWS CLI authentication



10. STEP 7: CONFIGURED AWS CLI

aws configure

- Region: ap-south-1
- Output: json



```
Mar 31 10:57:04 ip-172-31-43-245.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Mar 31 10:57:05 ip-172-31-43-245.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Mar 31 10:57:05 ip-172-31-43-245.ap-south-1.compute.internal httpd[25704]: Server configured, listening on: port 80

[ec2-user@ip-172-31-43-245 ~]$ aws --version
aws-cli/2.33.15 pythom/3.9.25 linux/x86_64.164-196.303.amzn2023.x86_64 source/x86_64.amzn.2023
[ec2-user@ip-172-31-43-245 ~]$ sudo yum install aws-cli -y
Last metadata expiration check: 0:04:44 ago on Tue Mar 31 10:55:58 2026.
Package awscli-2-2.33.15-1.amzn2023.0.1.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-43-245 ~]$ aws configure

[!]+ Stopped aws configure
[ec2-user@ip-172-31-43-245 ~]$ aws configure

[ec2-user@ip-172-31-43-245 ~]$ aws configure
AWS Access Key ID [None]: AKIAV6URMTPKK7YNN3W
AWS Secret Access Key [None]: nGwBYnK08zxWkUGXDW5AbJrdVN1X24cou/mViUq/
Default region name [None]: ap-south-1
Default output format [None]: [ec2-user@ip-172-31-43-245 ~]$
[ec2-user@ip-172-31-43-245 ~]$
[ec2-user@ip-172-31-43-245 ~]$ aws configure
[ec2-user@ip-172-31-43-245 ~]$ aws configure

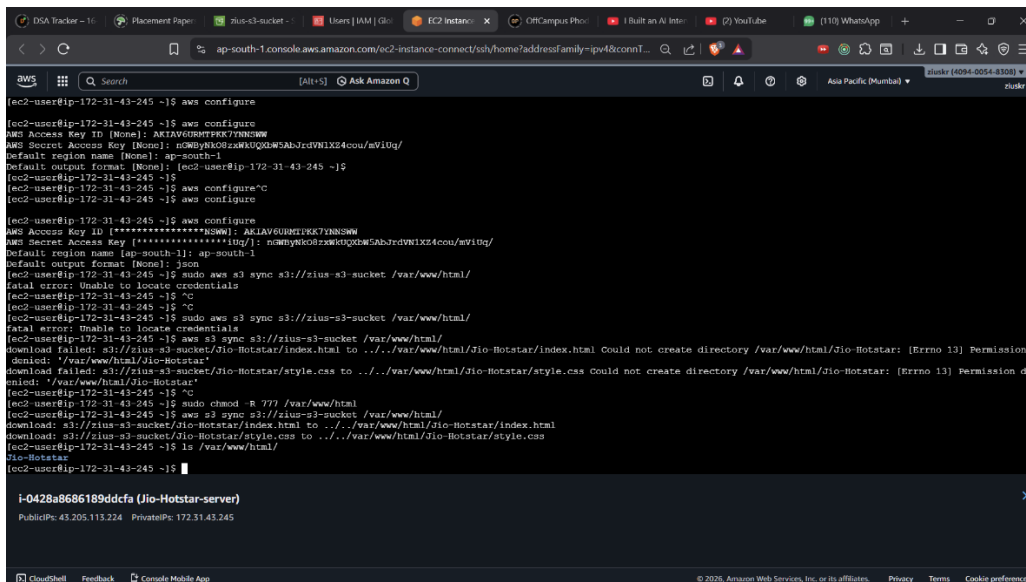
[ec2-user@ip-172-31-43-245 ~]$ aws configure
AWS Access Key ID [*****]: AKIAV6URMTPKK7YNN3W
AWS Secret Access Key [*****]: nGwBYnK08zxWkUGXDW5AbJrdVN1X24cou/mViUq/
Default region name [ap-south-1]: ap-south-1
Default output format [None]: json
[ec2-user@ip-172-31-43-245 ~]$
```

i-0428a8686189ddcfa (Jio-Hotstar-server)
PublicIPs: 43.205.113.224 PrivateIPs: 172.31.43.245

11. STEP 8: FIXED PERMISSIONS ISSUE

sudo chmod -R 777 /var/www/html

Allowed file transfer to web directory



```
[ec2-user@ip-172-31-43-245 ~]$ aws configure

[ec2-user@ip-172-31-43-245 ~]$ aws configure
AWS Access Key ID [None]: AKIAV6URMTPKK7YNN3W
AWS Secret Access Key [None]: nGwBYnK08zxWkUGXDW5AbJrdVN1X24cou/mViUq/
Default region name [None]: ap-south-1
Default output format [None]: [ec2-user@ip-172-31-43-245 ~]$
[ec2-user@ip-172-31-43-245 ~]$
[ec2-user@ip-172-31-43-245 ~]$ aws configure
[ec2-user@ip-172-31-43-245 ~]$ aws configure

[ec2-user@ip-172-31-43-245 ~]$ aws configure
AWS Access Key ID [*****]: AKIAV6URMTPKK7YNN3W
AWS Secret Access Key [*****]: nGwBYnK08zxWkUGXDW5AbJrdVN1X24cou/mViUq/
Default region name [ap-south-1]: ap-south-1
Default output format [None]: json
[ec2-user@ip-172-31-43-245 ~]$ sudo aws s3 sync s3://zius-s3-sucket /var/www/html/
fatal error: Unable to locate credentials
[ec2-user@ip-172-31-43-245 ~]$ ^C
[ec2-user@ip-172-31-43-245 ~]$ sudo aws s3 sync s3://zius-s3-sucket /var/www/html/
fatal error: Unable to locate credentials
[ec2-user@ip-172-31-43-245 ~]$ aws s3 sync s3://zius-s3-sucket /var/www/html/
download failed: s3://zius-s3-sucket/jio-Hotstar/index.html to ../var/www/html/jio-Hotstar/index.html: [Errno 13] Permission denied: '/var/www/html/jio-Hotstar'
download failed: s3://zius-s3-sucket/jio-Hotstar/style.css to ../var/www/html/jio-Hotstar/style.css: [Errno 13] Permission denied: '/var/www/html/jio-Hotstar'
[ec2-user@ip-172-31-43-245 ~]$ sudo chmod -R 777 /var/www/html
[ec2-user@ip-172-31-43-245 ~]$ aws s3 sync s3://zius-s3-sucket /var/www/html/
download s3://zius-s3-sucket/jio-Hotstar/index.html to ../var/www/html/jio-Hotstar/index.html
download s3://zius-s3-sucket/jio-Hotstar/style.css to ../var/www/html/jio-Hotstar/style.css
[ec2-user@ip-172-31-43-245 ~]$ ls /var/www/html/
jio-Hotstar
[ec2-user@ip-172-31-43-245 ~]$
```

i-0428a8686189ddcfa (Jio-Hotstar-server)
PublicIPs: 43.205.113.224 PrivateIPs: 172.31.43.245

12. STEP 9: SYNCED FILES FROM S3 TO EC2

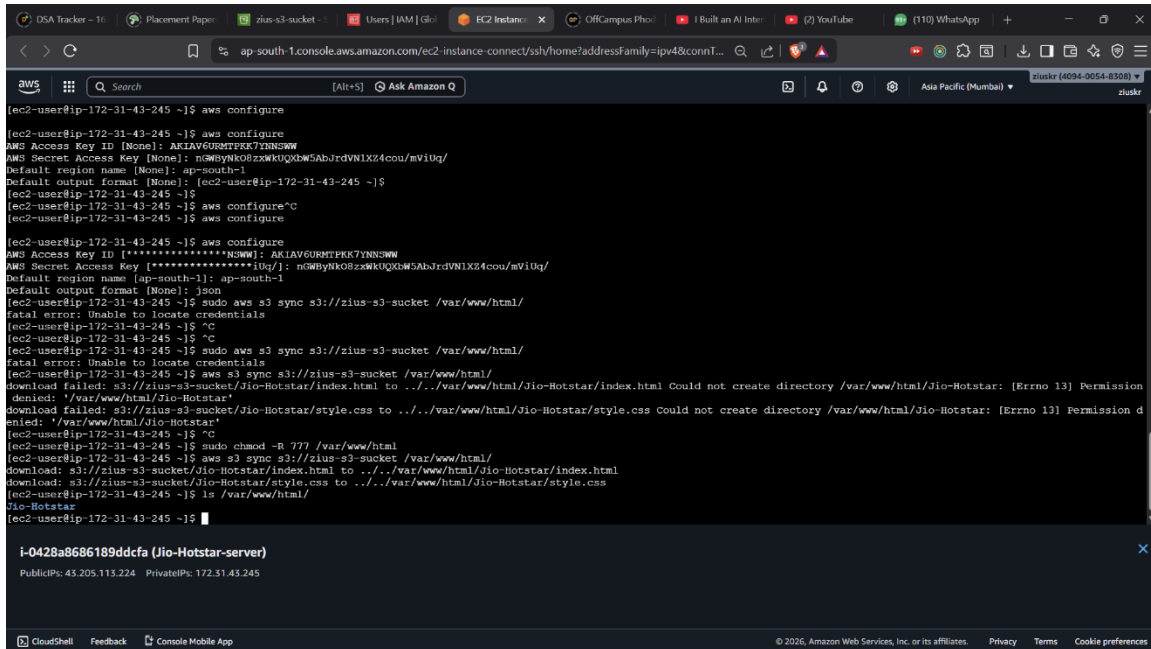
```
aws s3 sync s3://zius-s3-sucket /var/www/html/
```

Files copied successfully

13. STEP 10: VERIFIED FILES

```
ls /var/www/html/
```

```
ls /var/www/html/Jio-Hotstar/
```



```
[ec2-user@ip-172-31-43-245 ~]$ aws configure
[ec2-user@ip-172-31-43-245 ~]$ aws configure
AWS Access Key ID [None]: AKIAV6URMTPKK7YNN5NW
AWS Secret Access Key [None]: nGwByNkO8zxWkUQkH5AbJrdVN1XZ4cou/mv1Uq/
Default region name [None]: ap-south-1
Default output format [None]: json
[ec2-user@ip-172-31-43-245 ~]$
[ec2-user@ip-172-31-43-245 ~]$ aws configure
[ec2-user@ip-172-31-43-245 ~]$ aws configure

[ec2-user@ip-172-31-43-245 ~]$ aws configure
AWS Access Key ID [*****]: AKIAV6URMTPKK7YNN5NW
AWS Secret Access Key [*****]: nGwByNkO8zxWkUQkH5AbJrdVN1XZ4cou/mv1Uq/
Default region name [ap-south-1]: ap-south-1
Default output format [None]: json
[ec2-user@ip-172-31-43-245 ~]$ sudo aws s3 sync s3://zius-s3-sucket /var/www/html/
fatal error: Unable to locate credentials
[ec2-user@ip-172-31-43-245 ~]$ ^C
[ec2-user@ip-172-31-43-245 ~]$ ^C
[ec2-user@ip-172-31-43-245 ~]$ sudo aws s3 sync s3://zius-s3-sucket /var/www/html/
fatal error: Unable to locate credentials
[ec2-user@ip-172-31-43-245 ~]$ aws s3 sync s3://zius-s3-sucket /var/www/html/
download failed: s3://zius-s3-sucket/Jio-Hotstar/index.html to ../../var/www/html/Jio-Hotstar/index.html Could not create directory /var/www/html/Jio-Hotstar: [Errno 13] Permission
denied: '/var/www/html/Jio-Hotstar'
download failed: s3://zius-s3-sucket/Jio-Hotstar/style.css to ../../var/www/html/Jio-Hotstar/style.css Could not create directory /var/www/html/Jio-Hotstar: [Errno 13] Permission d
enied: '/var/www/html/Jio-Hotstar'
[ec2-user@ip-172-31-43-245 ~]$ sudo chmod -R 777 /var/www/html
[ec2-user@ip-172-31-43-245 ~]$ aws s3 sync s3://zius-s3-sucket /var/www/html/
download: s3://zius-s3-sucket/Jio-Hotstar/index.html to ../../var/www/html/Jio-Hotstar/index.html
download: s3://zius-s3-sucket/Jio-Hotstar/style.css to ../../var/www/html/Jio-Hotstar/style.css
[ec2-user@ip-172-31-43-245 ~]$ ls /var/www/html/
Jio-Hotstar
[ec2-user@ip-172-31-43-245 ~]$
```

i-0428a8686189ddcfa (Jio-Hotstar-server)

PublicIPs: 43.205.113.224 PrivateIPs: 172.31.43.245

CloudShell Feedback Console Mobile App

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14. STEP 11: MOVED FILES FOR HOSTING

```
sudo cp -r /var/www/html/Jio-Hotstar/* /var/www/html/
```

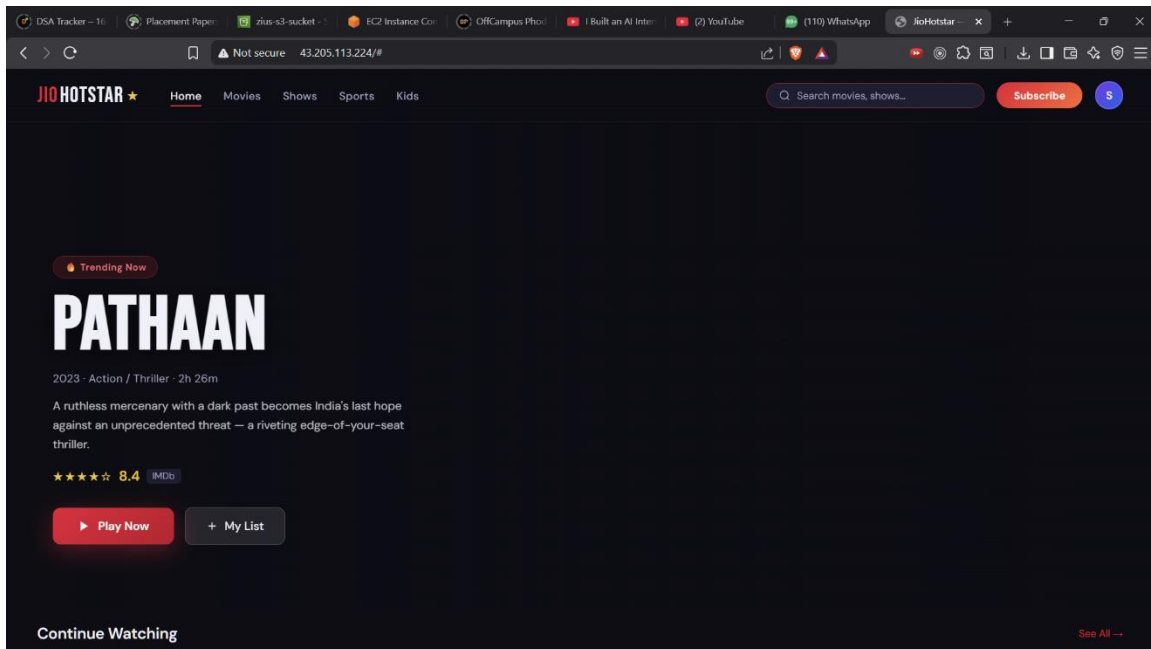
15. STEP 12: WEBSITE HOSTED

- Accessed via:

<http://43.205.113.224>

16. Video Of Backend Working

https://drive.google.com/file/d/1_E4StpZjAhwfVVECvHQND3nrMKNmytF-/view?usp=drive_link



Conclusion

The project successfully demonstrated deployment of a static website using AWS cloud services. It provided practical understanding of cloud storage, server configuration, access management, and web hosting. The integration of S3 and EC2 enabled efficient and scalable deployment.